Practical 9

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
   int source_fd, dest_fd;
   char buffer[BUFFER_SIZE];
   ssize_t bytes_read, bytes_written;

   if (argc != 3)
   {
      printf("Usage: %s <source> <destination>\n", argv[0]);
      return 1;
   }

   source_fd = open(argv[1], O_RDONLY);

   if (source_fd < 0)
   {
      perror("open source");
      return 1;
   }
```

```
dest_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

if (dest_fd < 0)
{
   perror("open destination");
   close(source_fd);
   return 1;
}

while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
{
   bytes_written = write(dest_fd, buffer, bytes_read);

   if (bytes_written != bytes_read)
   {
      perror("write");
      close(source_fd);
      close(dest_fd);
      return 1;
   }
}

if (bytes_read < 0)
   perror("read");

/* Demonstrate lseek() */
if (lseek(source_fd, 0, SEEK_SET) < 0)
   perror("lseek");
```

```
    close(source_fd);
    close(dest_fd);

    printf("File copied successfully.\n");

    return 0;
}
```

```
snigdha@Snigdha:~/quotes_practical$ nano source.txt
snigdha@Snigdha:~/quotes_practical$ nano filecopy.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o filecopy filecopy.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o filecopy filecopy.c
snigdha@Snigdha:~/quotes_practical$ ./filecopy source.txt destination.txt
File copied successfully.

snigdha@Snigdha:~/quotes_practical$
snigdha@Snigdha:~/quotes_practical$ cat destination.txt
This is a sample file.
This file is used to demonstrate Linux file I/O system calls.
snigdha@Snigdha:~/quotes_practical$ cmp source.txt destination.txt
snigdha@Snigdha:~/quotes_practical$ ls -l source.txt destination.txt
-rw-r--r-- 1 snigdha snigdha 85 Oct  7 03:23 destination.txt
-rw-r--r-- 1 snigdha snigdha 85 Oct  7 03:22 source.txt
```

```
#include <stdio.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    FILE *source, *destination;
    char buffer[BUFFER_SIZE];
    size_t bytes_read;
```

```
if (argc != 3)
{
   printf("Usage: %s <source> <destination>\n", argv[0]);
   return 1;
}

source = fopen(argv[1], "rb");

if (source == NULL)
{
   perror("fopen source");
   return 1;
}

destination = fopen(argv[2], "wb");

if (destination == NULL)
{
   perror("fopen destination");
   fclose(source);
   return 1;
}

while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, source)) > 0)
{
   fwrite(buffer, 1, bytes_read, destination);
}
```

```
    fclose(source);
    fclose(destination);

    printf("File copied using standard library functions.\n");

    return 0;
}
```

```
snigdha@Snigdha:~/quotes_practical$ nano filecopy_stdio.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o filecopy_stdio file
copy_stdio.c
snigdha@Snigdha:~/quotes_practical$ ./filecopy_stdio source.txt destination_
stdio.txt
File copied using standard library functions.
snigdha@Snigdha:~/quotes_practical$ cmp source.txt destination_stdio.txt
snigdha@Snigdha:~/quotes_practical$ time ./filecopy source.txt destination1.
txt
File copied successfully.

real    0m0.001s
user    0m0.001s
sys     0m0.000s
snigdha@Snigdha:~/quotes_practical$ time ./filecopy source.txt destination2.
txt
File copied successfully.

real    0m0.002s
user    0m0.002s
sys     0m0.000s
snigdha@Snigdha:~/quotes_practical$ dd if=/dev/zero of=largefile.bin bs=1M c
ount=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.0826066 s, 1.3 GB/s
snigdha@Snigdha:~/quotes_practical$ time ./filecopy largefile.bin copy_lowle
vel.bin
File copied successfully.

real    0m0.187s
user    0m0.016s
sys     0m0.168s
snigdha@Snigdha:~/quotes_practical$ time ./filecopy_stdio largefile.bin copy
_stdio.bin
File copied using standard library functions.

real    0m0.139s
user    0m0.012s
sys     0m0.127s
```

```
#include <stdio.h>
#include <fcntl.h>
```

```
#include <unistd.h>
#include <stdlib.h>

int main()
{
   int input_fd, output_fd;

   input_fd = open("input.txt", O_RDONLY);

   if (input_fd < 0)
   {
      perror("input.txt");
      return 1;
   }

   output_fd = open("output.txt",
               O_WRONLY | O_CREAT | O_TRUNC,
               0644);

   if (output_fd < 0)
   {
      perror("output.txt");
      close(input_fd);
      return 1;
   }

   /* Redirect standard input */
   if (dup2(input_fd, STDIN_FILENO) < 0)
```

```
    {
        perror("dup2 stdin");
        return 1;
    }

    /* Redirect standard output */
    if (dup2(output_fd, STDOUT_FILENO) < 0)
    {
        perror("dup2 stdout");
        return 1;
    }

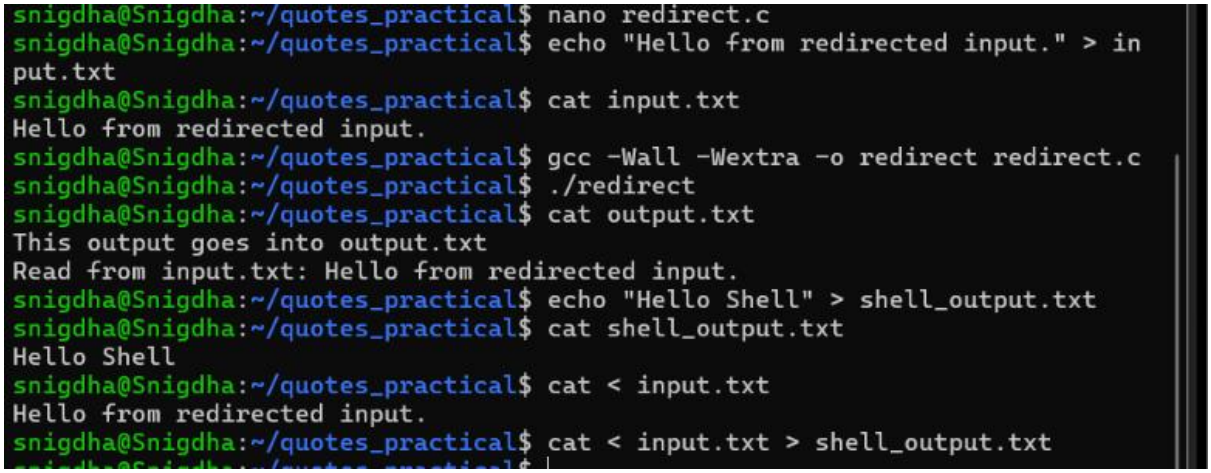
    close(input_fd);
    close(output_fd);

    printf("This output goes into output.txt\n");

    char buffer[100];

    if (fgets(buffer, sizeof(buffer), stdin) != NULL)
    {
        printf("Read from input.txt: %s", buffer);
    }

    return 0;
}
```


snigdha@Snigdha:~/quotes_practical$ nano redirect.c
snigdha@Snigdha:~/quotes_practical$ echo "Hello from redirected input." > in
put.txt
snigdha@Snigdha:~/quotes_practical$ cat input.txt
Hello from redirected input.
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o redirect redirect.c
snigdha@Snigdha:~/quotes_practical$ ./redirect
snigdha@Snigdha:~/quotes_practical$ cat output.txt
This output goes into output.txt
Read from input.txt: Hello from redirected input.
snigdha@Snigdha:~/quotes_practical$ echo "Hello Shell" > shell_output.txt
snigdha@Snigdha:~/quotes_practical$ cat shell_output.txt
Hello Shell
snigdha@Snigdha:~/quotes_practical$ cat < input.txt
Hello from redirected input.
snigdha@Snigdha:~/quotes_practical$ cat < input.txt > shell_output.txt